SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0604 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I Sanjay S/192521058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sanjay S

1.With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process.

1. Conceptual Understanding

A Machine Learning (ML) system enables computers to learn patterns from data and make predictions or discover useful information. The main functional blocks are data collection, data preprocessing, feature extraction, model training, model evaluation, and prediction.

1. Data Collection

Gather raw data

2. Data Preprocessing

Clean and prepare data

3. Feature Extraction

Identify important attributes

4. Model Training

Learn patterns using an algorithm

5. Model Evaluation

Check model performance

6. Prediction / Results

Predict outcomes or discover groups

2. Accuracy of Answer

Explanation of the functional blocks:

1. Data Collection: Collects information from databases, sensors, surveys, and websites.
2. Data Preprocessing: Handles missing values, duplicate records, incorrect data, and inconsistent formats.
3. Feature Extraction: Identifies useful characteristics from raw data for learning.
4. Model Training: Applies an algorithm to learn patterns from the prepared data.
5. Model Evaluation: Tests performance using suitable metrics such as accuracy or a silhouette score.
6. Prediction: Uses the trained model to predict outputs or identify groups in new data.

Supervised and unsupervised learning

Supervised learning	**Unsupervised learning**
Uses labelled data.	Uses unlabelled data.
Correct outputs are provided during training.	Correct outputs are not provided during training.
Learns to predict outputs.	Learns to discover patterns or groups.
Algorithms: Decision Tree, Linear Regression.	Algorithms: K-Means, Hierarchical Clustering.

3. Application of Knowledge

Example 1: Supervised learning – Spam email detection

- Collect emails labelled as Spam or Not Spam.
- Preprocess the text and extract features such as suspicious words.
- Train a classification model using the labelled emails.
- Evaluate the model using separate test emails.
- Predict whether a new email is Spam or Not Spam.

Example 2: Unsupervised learning – Customer segmentation

- Collect customer data such as spending amount and purchase frequency without group labels.
- Clean and scale the data.
- Extract relevant features.
- Apply the K-Means clustering algorithm.
- Evaluate the quality of the resulting clusters.
- Group customers into categories such as low-spending and high-spending customers.

4. Completeness of Response

The following points summarize how the two learning approaches handle data:

Aspect	**Supervised**	**Unsupervised**
Input data	Features with known labels	Features without labels
Learning process	Compares predictions with known answers	Finds similarities and hidden structures
Main tasks	Classification and regression	Clustering and dimensionality reduction
Evaluation	Accuracy, precision, recall, etc.	Silhouette score and cluster quality

Application	Disease prediction	Customer grouping

5. Clarity & Presentation

Conclusion: A machine learning system transforms raw data into useful predictions or patterns through a sequence of functional blocks. Supervised learning uses labelled data to predict known outputs, while unsupervised learning uses unlabelled data to discover hidden patterns. Both approaches support practical applications in healthcare, banking, business, and email filtering.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response.

1. Conceptual Understanding

A Machine Learning (ML) Disaster Management System uses historical and real-time data to estimate disaster risks, identify dangerous conditions, and support early-warning decisions for events such as floods and earthquakes.

The system collects information from weather stations, rainfall sensors, river-level sensors, satellite images, seismic sensors, and historical disaster records. It processes the data, trains suitable ML models, and provides risk estimates to disaster management authorities.

Neat sketch / block diagram

1. Data Collection

Historical records + real-time sensor, weather and satellite data

2. Data Preprocessing

Clean missing values, remove errors, synchronize timestamps

3. Feature Selection

Rainfall, river level, soil moisture, seismic signals and location

4. Model Training

Train suitable flood-risk and earthquake-related models

5. Validation and Evaluation

Test accuracy, recall, false alarms and warning lead time

6. Risk Assessment and Early Warning

Generate risk levels, alerts, evacuation support and response plans

2. Accuracy of Answer: Selection of an Appropriate Model

Different algorithms are suitable for different disaster types. A single model is not necessarily appropriate for every disaster.

Algorithm	Application	Justification
Decision Tree / Random Forest	Flood-risk prediction	Handles nonlinear relationships among rainfall, river levels, soil moisture, and terrain.
Neural Network (LSTM)	Flood forecasting using time-series data	Learns temporal patterns in rainfall, river levels, and other sequential measurements.
Convolutional Neural Network (CNN)	Satellite image analysis	Extracts spatial features from images to identify inundated areas or flood-prone regions.
Seismic signal analysis and statistical models	Earthquake monitoring	Helps detect and classify seismic signals and identify patterns associated with ground motion.

Justification: Random Forest is a good baseline for flood-risk classification because it is relatively easy to train and interpret. LSTM networks are suitable when sufficient sequential data are available. CNNs are useful for image-based analysis.

Earthquakes cannot currently be predicted reliably in terms of their exact time, location, and magnitude. ML can help detect seismic events, estimate shaking or damage risk, and support earthquake early-warning systems after an earthquake begins, but it cannot guarantee advance prediction.

3. Application of Knowledge: Data Preprocessing and Feature Selection

A. Data preprocessing

1. Data collection: Gather historical rainfall, river levels, flood records, seismic measurements, satellite images, and real-time sensor readings.
2. Missing-value handling: Fill suitable missing numerical values using interpolation or imputation, or flag unreliable records.
3. Noise removal: Filter sensor noise and remove duplicate or incorrect readings.
4. Data integration: Combine sources using consistent timestamps, geographic coordinates, and measurement units.
5. Normalization: Scale numerical features when required by the chosen algorithm.
6. Data labelling: For supervised flood models, label past observations using verified flood events or risk categories. For seismic models, use validated event catalogues and signal labels.
7. Class imbalance handling: Use class weights or careful resampling if severe disasters are much rarer than normal conditions.

B. Feature selection

Important features include:

- Floods: Rainfall intensity, accumulated rainfall, river water level, soil moisture, slope, drainage capacity, and historical flood frequency.

- Earthquakes: Seismic waveform characteristics, ground acceleration, signal amplitude, frequency content, station location, and measured ground motion.

Feature selection techniques include:

- Correlation analysis: Identifies strongly related variables.
- Feature importance: Uses Random Forest or similar models to rank useful features.
- Mutual information: Measures how informative a feature is about the target.
- Domain knowledge: Includes scientifically meaningful variables selected with disaster-management experts.

These steps reduce irrelevant information, improve model efficiency, and may increase prediction performance.

4. Completeness of Response: Training, Validation, and Evaluation

Step 1: Training

Divide the historical dataset into training, validation, and test sets. Train the selected algorithm using past observations and their verified outcomes.

For time-series data, use chronological splitting rather than random splitting to avoid future information leaking into the past.

Step 2: Validation

Use validation data to select model parameters, compare algorithms, and choose appropriate warning thresholds. Check performance across different locations and seasons.

Step 3: Evaluation

Evaluate the final model on unseen test data.

Metric	Purpose
Accuracy	Measures the proportion of correct predictions.
Precision	Measures how many issued positive predictions were correct.
Recall	Measures how many actual disaster events were detected.
F1-score	Balances precision and recall.
False-alarm rate	Helps assess unnecessary warnings.
Warning lead time	Measures how early a useful warning is available.

For disaster management, recall is especially important because missing a dangerous event can cause serious harm. However, false alarms must also be controlled to maintain public trust.

Step 4: Real-time prediction

Feed current sensor readings and updated weather information into the deployed model. The system generates risk estimates, displays them on a monitoring dashboard, and sends alerts when validated thresholds are crossed.

Step 5: Continuous monitoring

Compare predictions with actual outcomes, monitor changes in sensor quality and model performance, and retrain the model when necessary using verified new data.

5. Clarity & Presentation: Role in Disaster Management

Early warning

Flood-risk forecasts can trigger alerts when rainfall and river conditions indicate danger. Earthquake early-warning systems can provide seconds of warning after an earthquake is detected, before stronger shaking reaches some locations.

Risk assessment

Authorities can identify vulnerable villages, low-lying regions, roads, hospitals, and buildings that may need priority protection.

Emergency response

Risk maps and alerts can help officials plan evacuations, allocate rescue teams, position relief supplies, and choose safer transport routes.

Conclusion: A machine learning disaster management system combines historical and real-time data, preprocessing, feature selection, suitable algorithms, and systematic evaluation to estimate disaster risks. Random Forest and neural networks can support flood forecasting, while seismic detection and early-warning techniques can help reduce earthquake impacts. When integrated with reliable sensors, official warning procedures, and emergency response teams, the system can improve preparedness and help protect lives and property.

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

1. Conceptual Understanding

A Multilayer Perceptron (MLP) is a feedforward neural network consisting of an input layer, one or more hidden layers, and an output layer. It is used for tasks such as binary classification, where the output belongs to one of two classes, such as Pass/Fail or Spam/Not Spam.

Backpropagation is a learning algorithm that calculates how much each weight contributes to the prediction error and adjusts the weights to reduce that error.

2. Accuracy of Answer: Steps in Backpropagation

Step 1: Forward pass

Input data passes through the network. Each neuron calculates a weighted sum and applies an activation function.

$$z = \sum_i w_i x_i + b$$

$$a = f(z)$$

Here, x_i represents inputs, w_i represents weights, b is the bias, and f is the activation function.

For binary classification, the output neuron commonly uses the sigmoid function:

$$\hat{y} = \frac{1}{1 + e^{-z}}$$

The predicted value $\hat{y}$ lies between 0 and 1.

Step 2: Error calculation

The predicted output is compared with the actual target \(y\). Binary cross-entropy is a common loss function:

$$L = -[y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})]$$

A higher loss indicates a worse prediction, while a lower loss indicates a better prediction.

Step 3: Backward propagation

The error is propagated backward from the output layer to the hidden layers using the chain rule of calculus. This calculates the gradient of the loss with respect to each weight.

For a weight w:

$$\frac{\partial L}{\partial w}$$

The gradient indicates how changing the weight affects the loss.

Step 4: Weight update

Weights are updated using gradient descent:

$$\boxed{w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}}$$

Here, \(\eta\) is the learning rate. Biases are updated similarly. The process repeats over many training examples and epochs until the model achieves satisfactory performance.

3. Application of Knowledge: Simple Numerical Example

Consider a binary classifier predicting whether a student passes an examination.

Assume a simplified network with one input, one sigmoid output neuron, input \(x=1\), initial weight \(w=0\), bias \(b=0\), target \(y=1\), and learning rate \(\eta=0.1\).

Step A — Forward pass

$$z = wx + b = (0)(1) + 0 = 0$$

$$\hat{y} = \frac{1}{1 + e^{0}} = 0.5$$

The model predicts a probability of 0.5.

Step B — Calculate error

Using binary cross-entropy:

$$L = -\ln(0.5) \approx 0.6931$$

The model has not yet learned to predict the correct class confidently.

Step C — Calculate the gradient

For a sigmoid output with binary cross-entropy:

$$\frac{\partial L}{\partial w} = (\hat{y} - y)x$$

$$= (0.5 - 1)(1) = -0.5$$

Step D — Update the weight

$$w_{\text{new}} = 0 - 0.1(-0.5) = 0.05$$

The bias is updated similarly:

$$b_{\text{new}} = 0 - 0.1(0.5 - 1) = 0.05$$

Step E — New prediction

$$z = (0.05)(1) + 0.05 = 0.1$$

$$\hat{y} = \frac{1}{1 + e^{-0.1}} \approx 0.525$$

The predicted probability increases from 0.5 to approximately 0.525, moving toward the target of 1.

This is one training update. An actual MLP repeats this process for all weights and biases across its hidden and output layers.

4. Completeness of Response: Learning Rate and Activation Functions

A. Effect of learning rate

The learning rate determines the size of each weight update.

Learning rate	**Effect**
Too small	Training is slow and convergence takes longer.
Too large	Updates may overshoot the minimum, causing unstable training or divergence.
Appropriate	Training generally progresses more smoothly toward a low-loss solution.

B. Effect of activation functions

Activation function	**Main use**	**Effect on learning**
Sigmoid	Binary output layer	Produces probabilities between 0 and 1; can suffer from vanishing gradients in hidden layers.
ReLU	Hidden layers	Helps reduce vanishing-gradient problems for positive activations and is computationally efficient.
Tanh	Hidden layers	Produces outputs between −1 and 1; can still suffer from

		vanishing gradients.
Softmax	Multiclass output layer	Produces class probabilities across multiple mutually exclusive classes.

For binary classification, a common combination is ReLU in hidden layers and sigmoid in the output layer, with binary cross-entropy as the loss function.

5. Clarity & Presentation: Conclusion

Backpropagation trains a multilayer perceptron by performing a forward pass, calculating the prediction loss, propagating gradients backward, and updating weights and biases using gradient descent. The learning rate controls the size of updates, while activation functions influence gradient flow and the network's ability to learn complex patterns. Choosing suitable settings helps improve convergence, prediction accuracy, and overall model performance.

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-bystep improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks) Dataset: Example Speed Accuracy Fitness Value Suitable Solution 1 High Low 40 No 2 Medium Medium 60 Yes 3 High Medium 75 Yes 4 Low High 65 Yes 5 Medium High 80 Yes

1. Conceptual Understanding

A Genetic Algorithm (GA) is an optimization technique inspired by natural selection and evolution. It improves candidate solutions over generations using three main operations:

1. Selection: Chooses the fitter candidate solutions as parents.
2. Crossover: Combines parts of two parents to create new solutions.
3. Mutation: Randomly changes a parameter to introduce diversity and possibly improve fitness.

The objective is to find the candidate solution with the highest fitness value.

2. Accuracy of Answer: Given Dataset

Example	Speed	Accuracy	Fitness	Suitable
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Initial best solution: Example 5, with Speed = Medium, Accuracy = High, and Fitness = 80.

3. Application of Knowledge: GA Operations

Step 1: Selection

Select fitter solutions as parents. For example:

- Parent 1: Example 3 — (High, Medium), Fitness = 75
- Parent 2: Example 5 — (Medium, High), Fitness = 80

These solutions are selected because they have high fitness values.

Step 2: Crossover

Exchange one parameter between the two parents.

Parent	Speed	Accuracy
Parent 1	High	Medium
Parent 2	Medium	High

After crossover, possible offspring are:

- Offspring 1: (High, High)
- Offspring 2: (Medium, Medium)

Assume the fitness of Offspring 1 is 90 and Offspring 2 is 60. These are illustrative fitness values because the dataset does not provide a formula for calculating fitness for new combinations.

Step 3: Mutation

Mutation randomly changes one parameter in an offspring.

For example:

- Before mutation: (High, Medium), Fitness = 75
- After mutation: (High, High), Fitness = 90

If the new combination receives a higher fitness, it becomes a stronger candidate for the next generation.

4. Completeness of Response: Improvement Across Generations

The following is an illustrative evolution based on the assumed fitness of 90 for the (High, High) solution.

Generation	Best candidate	Fitness
0 — Initial population	(Medium, High)	80
1 — After selection and crossover	(High, High)	90
2 — After further evaluation	(High, High)	90

The algorithm retains the best candidate and continues searching until a stopping condition is reached, such as a maximum number of generations or no further improvement.

5. Clarity & Presentation: Optimal Solution

Best optimal candidate in this illustrative example:

- Speed: High
- Accuracy: High
- Fitness value: 90
- Suitable: Yes, assuming the fitness threshold is met.

Conclusion: Genetic Algorithms improve candidate solutions through selection, crossover, and mutation. In the given dataset, Example 5 is the best initial solution with fitness 80. By combining and modifying candidate solutions, the algorithm may discover a better solution such as (High, High). However, its fitness of 90 is assumed for illustration; the actual optimum requires a defined fitness function for every Speed–Accuracy combination.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)Dataset: Example Age Blood Pressure Glucose Level Lifestyle Risk Actual Output Predicted Output 1 45 High High High Yes Yes 2 50 Normal High Medium Yes No 3 30 Normal Normal Low No No 4 60 High High High Yes Yes 5 40 Normal Normal Medium No Yes Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

1. Conceptual Understanding

A Neural Network is a machine learning model that learns patterns from input data to predict an output. In this problem, the model predicts whether a patient has a disease (Yes) or does not have a disease (No) using Age, Blood Pressure, Glucose Level, and Lifestyle Risk.

The network is trained using backpropagation, which adjusts weights and biases to minimize prediction errors.

Neural network diagram

Input Layer

Age • Blood Pressure • Glucose Level • Lifestyle Risk

Hidden Layer

Learns patterns using weights and activation functions

Output Layer

Disease: Yes / No

2. Accuracy of Answer: Training Using Backpropagation

The neural network is trained through the following steps:

1. Input: Provide the patient's age, blood pressure, glucose level, and lifestyle risk.
2. Preprocessing: Encode categorical values and scale numerical features.

3. Forward propagation: Calculate the predicted disease probability using weights, biases, and activation functions.
4. Error calculation: Compare the prediction with the actual output using a loss function such as binary cross-entropy.
5. Backpropagation: Calculate gradients of the loss with respect to the weights and biases.
6. Weight update: Apply gradient descent to reduce the loss.
7. Repeat: Continue training over multiple epochs until performance becomes satisfactory.

The weight update rule is:

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}$$

Here, w is the weight,η is the learning rate, and L is the loss.

3. Application of Knowledge: Confusion Matrix

Compare the actual outputs with the predicted outputs.

Example	Actual	Predicted	Result
1	Yes	Yes	True Positive (TP)
2	Yes	No	False Negative (FN)
3	No	No	True Negative (TN)
4	Yes	Yes	True Positive (TP)
5	No	Yes	False Positive (FP)

Taking Yes as the positive class:

- True Positives (TP) = 2
- True Negatives (TN) = 1
- False Positives (FP) = 1
- False Negatives (FN) = 1

Confusion matrix

Actual / Predicted	Predicted Yes	Predicted No
Actual Yes	TP = 2	FN = 1
Actual No	FP = 1	TN = 1

4. Completeness of Response: Evaluation Metrics

Total number of examples:

$$N = TP + TN + FP + FN = 2 + 1 + 1 + 1 = 5$$

1. Accuracy

$$\text{Accuracy} = \frac{TP + TN}{N} \times 100$$

$$= \frac{2 + 1}{5} \times 100 = \boxed{60\%}$$

2. Precision

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

$$= \frac{2}{2 + 1} \times 100 = \boxed{66.67\%}$$

3. Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN} \times 100$$

$$= \frac{2}{2 + 1} \times 100 = \boxed{66.67\%}$$

4. Specificity

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100$$

$$= \frac{1}{1 + 1} \times 100 = \boxed{50\%}$$

5. F1-score

$$F1 = \frac{2TP}{2TP + FP + FN}$$

$$= \frac{4}{4 + 1 + 1} \times 100 = \boxed{66.67\%}$$

Performance summary

Metric	**Result**
Accuracy	60%
Precision	66.67%
Recall	66.67%
Specificity	50%
F1-score	66.67%

The model correctly classifies 3 out of 5 patients. It misses one patient who has the disease and incorrectly predicts disease in one patient who does not have it. These results indicate that the model needs improvement. Since this is a very small dataset, the metrics are only illustrative and cannot establish reliable clinical performance.

5. Clarity & Presentation: Preprocessing and Feature Selection

Preprocessing improves reliability by:

- Handling missing values and duplicate records.
- Scaling Age and other numerical features where appropriate.
- Encoding Blood Pressure and Lifestyle Risk into numerical values.
- Checking inconsistent or incorrect measurements.
- Splitting data into training, validation, and test sets without data leakage.

Feature selection improves accuracy by:

- Identifying relevant features such as Glucose Level and Blood Pressure.
- Removing irrelevant or redundant attributes.
- Reducing model complexity and the risk of overfitting.
- Using domain knowledge and feature-importance analysis to retain useful predictors.

Conclusion: A neural network uses backpropagation to learn from patient data and classify disease presence. For the given dataset, TP = 2, TN = 1, FP = 1, and FN = 1, resulting in 60% accuracy. Appropriate preprocessing, feature selection, and evaluation on a larger independent dataset are necessary before considering any real medical use.